

LAKSHYA

NEET 2027

Lecture No.- 7

Physics

Electric Charges and
Fields

By – Manish Raj (MR Sir)



Today's Goal
→ Superposition theorem.
3 Questions on Coloumbs law

● aaT ki class rough Page pe Question solve kar ke
karna hai,

⇒ After write those Question Marked in ppt

Coulombs Law

$$F = k \frac{q_1 q_2}{r^2} \checkmark$$

magnitude of force

① $F_{12} = F_{21}$

② central

③ valid for point charge & spherical charge

④ same charge $\rightarrow$ repulsion
opposite $\rightarrow$ attract

⑤ conservative

⑥ long range

⑦ Inverse square force $F \propto \frac{1}{r^2}$



⑧ force may be attract and repulse

⑨ force on charge depend on medium

⑩ $F_{air} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$ $k = \left(\frac{1}{4\pi\epsilon_0}\right) = 9 \times 10^9$

$F_{med} = \frac{q_1 q_2}{4\pi\epsilon_m r^2}$ $\epsilon_r(\frac{1}{k}) = \frac{\epsilon_m}{\epsilon_0}$

$F_{med} = \frac{q_1 q_2}{4\pi\epsilon_r \epsilon_0 r^2} = \frac{F_{air}}{\epsilon_r} = \frac{F_{air}}{\epsilon_r}$ $\epsilon_m = \epsilon_r \epsilon_0$

Net electro. $\Rightarrow$ ⑪ electrostatic force mediated Photon

⑧ ^{force} charge $-4C$ and $+12C$ on two small sphere at distance x is F
now each charge given $+4C$ charge, then force between them will be y

Solⁿ

$$-4C + 4C = 0$$

$$+12C + 4C = 16C$$



$$F' = \underline{\underline{zero}}$$



Question

copy



Find ratio of the electrostatics and gravitational force between two protons which is at distance r .

$$m_p = 1.67 \times 10^{-27}$$

$$e = 1.6 \times 10^{-19}$$

$$\frac{F_{\text{electr}}}{F_{\text{grav}}} \approx 10^{+36}$$

$$\frac{F_{\text{electr}}}{F_{\text{grav}}} = \frac{\frac{K e^2}{r^2}}{\frac{G m_p^2}{r^2}}$$

$$= \frac{9 \times 10^9 (1.6 \times 10^{-19})^2}{6.67 \times 10^{-11} \times (1.67 \times 10^{-27})^2}$$

$$= \frac{9 \times 10^9 \times 10^{-38}}{6.67 \times 10^{-11} \times 10^{-54}}$$

$$= \frac{1.2 \times 10^{25}}{10^{-11}}$$

$$= 1.2 \times 10^{36}$$

copy

for Two Proton

$$\frac{F_{\text{electrostatic}}}{F_{\text{gravitation}}} = 10^{+36}$$

for Two Electron at distⁿ (r)

$$\frac{F_{\text{electrostatic}}}{F_{\text{gravitation}}} \approx 10^{+42}$$

*

for e & Proton.

$$\frac{F_{\text{gravitation}}}{F_{\text{electrostatic}}} = 10^{-39}$$

Question

H/w (1)

No copy



Two charge of 1 nC are at distance 3 cm apart in vacuum, then the force between them is?

$$1 \text{ nC} = 10^{-9} \text{ C}$$

$$10^{-9} \text{ C} \quad \text{-----} \quad 3 \text{ cm} \quad \text{-----} \quad 10^{-9} \text{ C}$$

$$F = \frac{9 \times 10^9 \times 10^{-9} \times 10^{-9}}{(3 \times 10^{-2})^2}$$

$$F = \frac{9 \times 10^{-9}}{9 \times 10^{-4}} = 10^{-5} \text{ N}$$



Q Two Identical point charge apply force on each other F ,
copy if 50% charge transferred from one to other then
 new force between them will be.

Solⁿ



$$F = \frac{K q \times q}{r^2} \quad \text{--- (1)}$$



$$F = \frac{K \frac{q}{2} \times \frac{3q}{2}}{r^2} = \frac{K \cancel{3} q \times q}{\cancel{4} r^2} = \frac{3}{4} F \quad \checkmark$$

% change in force

$$= \frac{f' - f_i}{f_i} \times 100$$

$$= \frac{\frac{3}{4} F - F}{F} \times 100$$

$$= \frac{3F - 4F}{4F} \times 100$$

$$= \frac{-\cancel{F}}{\cancel{4} F} \times 100 = -25\%$$

Question

② No copy



Two point charges A and B, having charges $+Q$ and $-Q$ respectively, are placed at certain distance apart and force acting between them is F. If 25% charge of A is transferred to B, then force between the charges becomes **[2019]**

1 $4F/3$

2 F

3 $9F/16$

4 $16F/9$



$$F = \frac{kQ \times Q}{r^2}$$

$$Q - 25\%Q = Q - \frac{25}{100}Q$$

$$= \frac{3Q}{4}$$

$$F' = \frac{k \cdot \frac{3Q}{4} \times Q}{r^2}$$

$$F' = \frac{9}{16} F$$

$$\left(\frac{3Q}{4}\right) \longrightarrow -Q + \frac{Q}{4} = \left(-\frac{3Q}{4}\right)$$

Question

③ copy



Two point charges repel each other with a force of 100 N. One of the charges is increased by 10% and other is reduced by 10%. Then new force of repulsion at the same distance would be:

- 1 99 N
- 2 121 N
- 3 100 N
- 4 110 N

$$F = \frac{K q^2}{r^2} = 100 \text{ N}$$

$$F' = K \frac{110\% q}{10} \times \frac{90\% q}{10} \frac{1}{r^2} = \frac{100 \text{ N} \times 99}{100} = 99 \text{ N}$$

Question

4 No copy



Two identical charged conducting spheres A and B have their centres separated by a certain distance. Charge on each sphere is q and the force of repulsion between them is F . A third identical uncharged conducting sphere is brought in contact with sphere A first and then with B and finally removed from both. New force of repulsion between spheres A and B (Radii of A and B are negligible compared to the distance of separation so that for calculating force between them they can be considered as point charges) is best given as:

[2025]

NEET

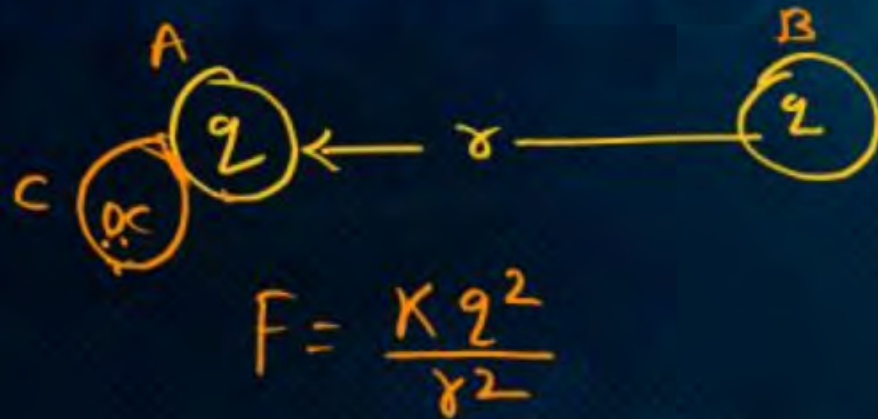
(No Induction)

1 $F/2$

2 $3F/8$

3 $3F/5$

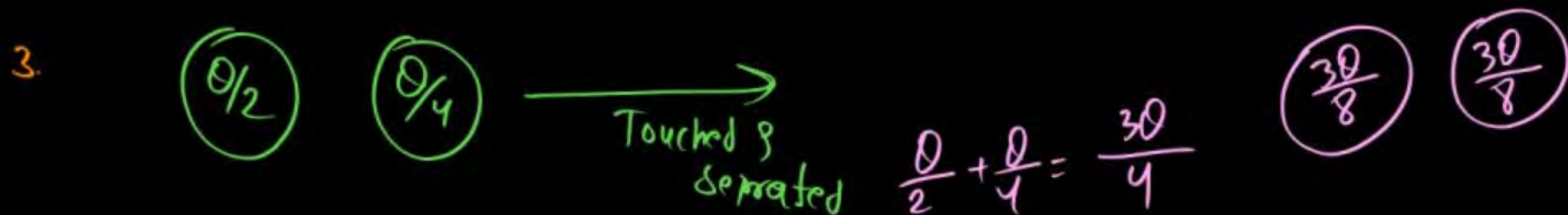
4 $2F/3$



$$F' = K \frac{q}{2} \times \frac{3q}{4} \frac{1}{r^2} = \frac{3Kq^2}{8r^2} = \frac{3}{8} F$$



$$q_{\text{total}} = 2 + \frac{2}{2} = \frac{32}{2}$$



When air is replaced by a ^{*}dielectric medium of constant K, the maximum force of attraction between two charges separated by a distance **[1999]**

- 1 increases K times
- 2 remains unchanged
- 3 decreases K times
- 4 increases K-1 times.

Question

(6) copy



Force between two point charges q , and q , placed in vacuum at ' r ' cm apart is F . Force between them when placed in a medium having dielectric $K = 5$ at ' $r/5$ ' cm apart will be:

[31 Jan, 2024 (Shift-II)]

1 $F/25$

2 $5F$ ✓

3 $F/5$ ✗

4 $25F$

$$F = \frac{Kq^2}{r^2}$$

$$F'_{\text{med}} = \frac{K \times q^2}{\left(\frac{r}{5}\right)^2 \times 5} = \frac{Kq^2}{\frac{r^2}{25} \times 5} = 5 \left(\frac{Kq^2}{r^2} \right)$$

$$F' = 5F$$

(Q) find "minimum electrostatic" force between charge placed at
1m distance.

copy

Solⁿ



$$F = \frac{kq^2}{r^2}$$

The equation is crossed out with a large 'X'.

$$F_{\min} = \frac{k(q_1 q_2)_{\min}}{(1)^2}$$

$$F_{\min} = k(e \times e)$$

$$= 9 \times 10^9 (1.6 \times 10^{-19})^2$$

$$= 9 \times 2.56 \times 10^9 \times 10^{-38} \text{ N}$$

Ans

Q Two Identical point charge (Q, m) placed at some distance if gravitational force is equal to electrostatic force then find $(Q/m) = ??$



$$\frac{Q}{m} = \sqrt{\frac{G}{K}} \quad \text{A}$$

$$= \sqrt{\frac{6.67 \times 10^{-11}}{9 \times 10^9}}$$

$$F_{\text{elect}} = F_{\text{gravit.}}$$

$$\frac{KQ^2}{r^2} = \frac{Gm^2}{r^2}$$

$$\frac{Q^2}{m^2} = \frac{G}{K}$$

(Q) Two identical charge body Apply force on each other is F at distance r then find total no of electron missing from the body.
COPY

Soln



$$q = ne \quad \text{--- (1)}$$

~~Zero~~

$$n = \frac{\sqrt{4\pi\epsilon_0 r^2 F}}{e}$$

Ans

$$F = \frac{kq^2}{r^2}$$

$$F = \frac{q^2}{4\pi\epsilon_0 r^2} = \frac{(ne)^2}{4\pi\epsilon_0 r^2}$$

$$(ne)^2 = 4\pi\epsilon_0 r^2 F$$

$$ne = \sqrt{4\pi\epsilon_0 r^2 F}$$

G.P series

No copy

$$a, ar, ar^2, ar^3, ar^4, \dots$$

$\uparrow$
1st term

$$r(\text{common ratio}) = \frac{n^{\text{th}} \text{ term}}{(n-1)^{\text{th}} \text{ term}}$$

$$a=2, r=3$$

Ex- 2, 6, (18), 54, 162, - -

$$a=2 \quad r = \frac{18}{6} = 3$$

no copy

Ex -

$$(Q) \quad 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots$$

$$a = 1 \quad r = \frac{\frac{1}{8}}{\frac{1}{4}} = \frac{1 \times 4}{8 \times 1} = \frac{1}{2}$$

$$\text{Sum of all term} = \frac{a}{1 - r(r)}$$

$$= \frac{1}{1 - \frac{1}{2}} = \frac{1}{\frac{1}{2}} = 2$$

copy

find force on q_0 charge which is at origin due to all other charges as shown in fig.

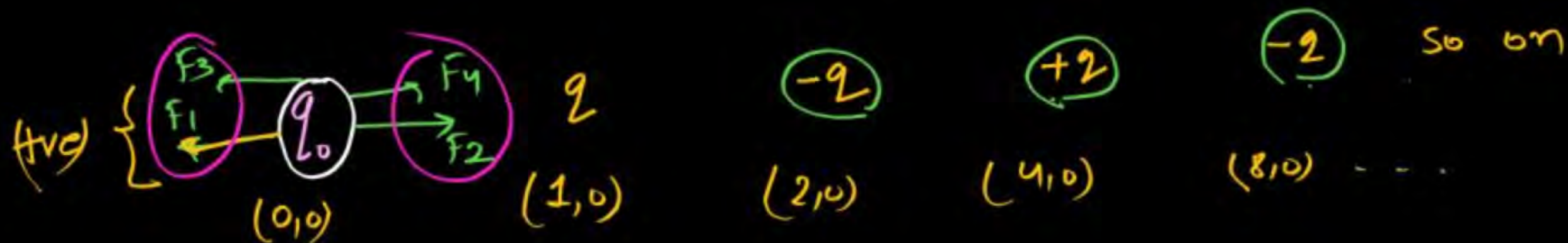


$$F_1 = \frac{K q_0 q}{(1)^2} \quad F_2 = \frac{K q_0 q}{(2)^2} \quad F_3 = \frac{K q_0 q}{(4)^2} \quad \dots \quad F_4 = \frac{K q_0 q}{(8)^2}$$

$$\begin{aligned} F_{\text{net on } q_0} &= \frac{K q_0 q}{1} + \frac{K q_0 q}{4} + \frac{K q_0 q}{16} + \dots \\ &= K q_0 q \left[1 + \frac{1}{4} + \frac{1}{16} + \dots \right] = K q_0 q \left[\frac{a}{1-r} \right] = K q_0 q \left[\frac{1}{1-\frac{1}{4}} \right] \\ &= K q_0 q \left(\frac{1}{\frac{3}{4}} \right) = \frac{4Kq}{3} \end{aligned}$$

No copy

force on q_0 charge due to all other charges



to 1740
one leg
stand

$$F_1 = \frac{Kq_0q}{1^2}$$

$$F_2 = \frac{Kq_0q}{(2)^2}$$

$$F_3 = \frac{Kq_0q}{4^2} \dots$$

~~Same~~

$$F_{net} = \frac{Kq_0q}{1} - \frac{Kq_0q}{4} + \frac{Kq_0q}{16} - \frac{Kq_0q}{64} \dots$$

$$a=1$$

$$r = \frac{-\frac{1}{4}}{1} = -\frac{1}{4}$$

$$= Kq_0q \left[1 - \frac{1}{4} + \frac{1}{16} - \frac{1}{64} + \dots \right]$$

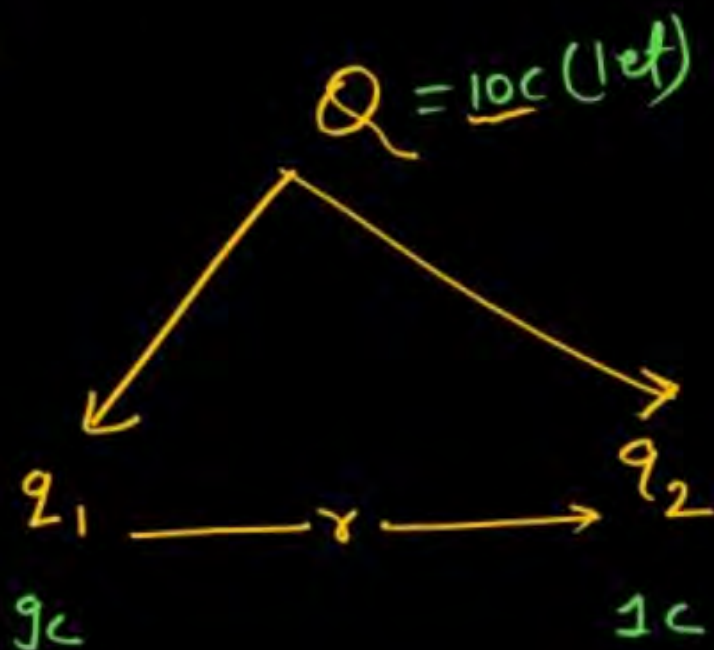
G.P. sum

$$F_{net} = Kq_0q \left[\frac{1}{1 - (-\frac{1}{4})} \right] = \frac{Kq_0q}{(1 + \frac{1}{4})} = \frac{4Kq_0q}{5}$$

copy

A Point charge Q divided into two Part q_1 and q_2 and Placed at some distance then find $\left(\frac{q_1}{q_2}\right)$ so that force between them will be maximum

Soln



$$q_1 + q_2 = Q = 10C$$

$$\frac{q_1}{q_2} = \frac{1}{1} \quad \text{max}$$

$$\frac{q_1}{Q} = \frac{1}{2}$$

$$F_{\text{max}} = \frac{K 5C \times 5C}{r^2}$$



$$F = \frac{K q_1 q_2}{r^2}$$

$$F = \frac{K 9C \times 1C}{r^2}$$

$$F = \frac{K 8C \times 2C}{r^2}$$

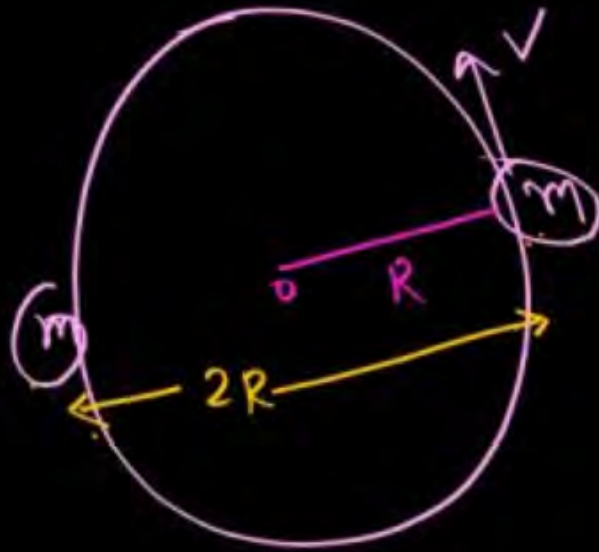
$$F = \frac{K 7C \times 3C}{r^2}$$

$$F = \frac{K 6C \times 4C}{r^2}$$

gravitation

140

Two Identical Point mass moving on circular Path
due to mutual gravitational force then
find their speed??

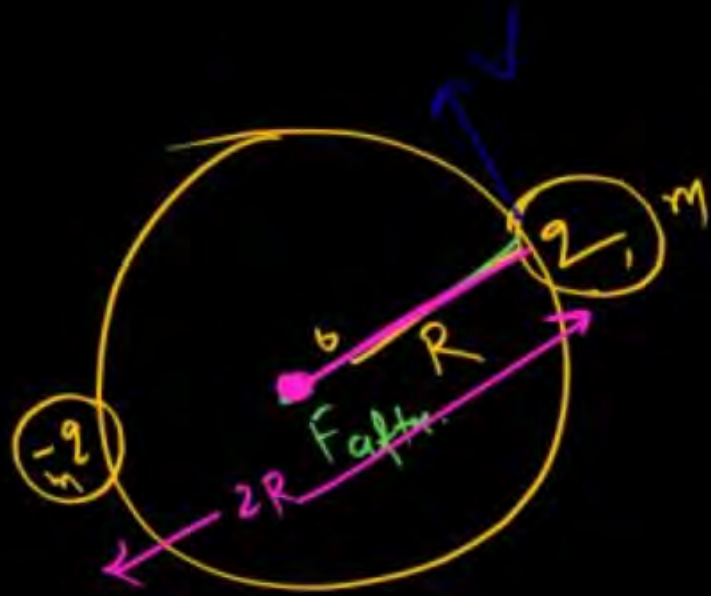


$$\frac{mv^2}{R} = \frac{Gm \times m}{(2R)^2}$$

$$\frac{v^2}{R} = \frac{Gm}{4R^2}$$

$$V = \sqrt{\frac{Gm}{4R}} \quad \underline{\text{Ans}}$$

copy Two point charge (q, m) and $(-q, m)$ are moving on circular path due to mutual electrostatic force then find their speed.



$$\left(\frac{mv^2}{R} \right) = \frac{Kq^2}{(2R)^2}$$

$$\frac{mv^2}{R} = \frac{Kq^2}{4R^2}$$

$$v^2 = \frac{Kq^2}{4mR}$$

$$v = \sqrt{\frac{Kq^2}{4mR}}$$

As

(Q) Two Identical Point charge apply force F when they are at distance γ . Now both are placed on two conducting spheres of Radius $[R < \gamma]$ and placed at same distance then force b/w them will be.

23.1

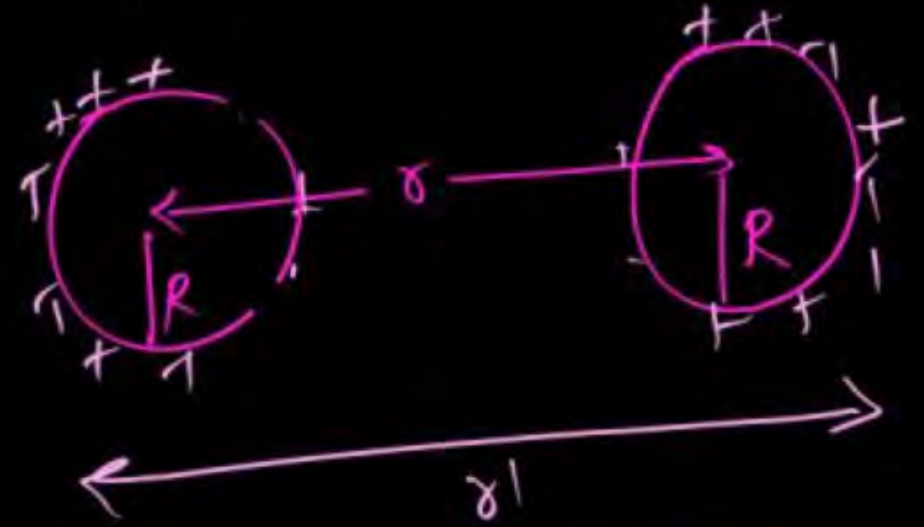
(a) $F' < F$

(b) $F' > F$

(c) $F' = F$



$$F = \frac{Kq^2}{\gamma^2}$$



$$F < F'$$

ans

Q Force between two identical point charge placed at distance x is F , now they placed on small sphere of radius R at distance x ($x \gg R$) then force between them.

1

2

$$F' = F \quad \underline{\underline{\text{same}}}$$



(Q) Two equal & opposite point charge apply force F when they are at distance x Now both are place on two conducting sphere of Radius $(R < x)$ and placed at same distance then force b/w them will be.

No copy



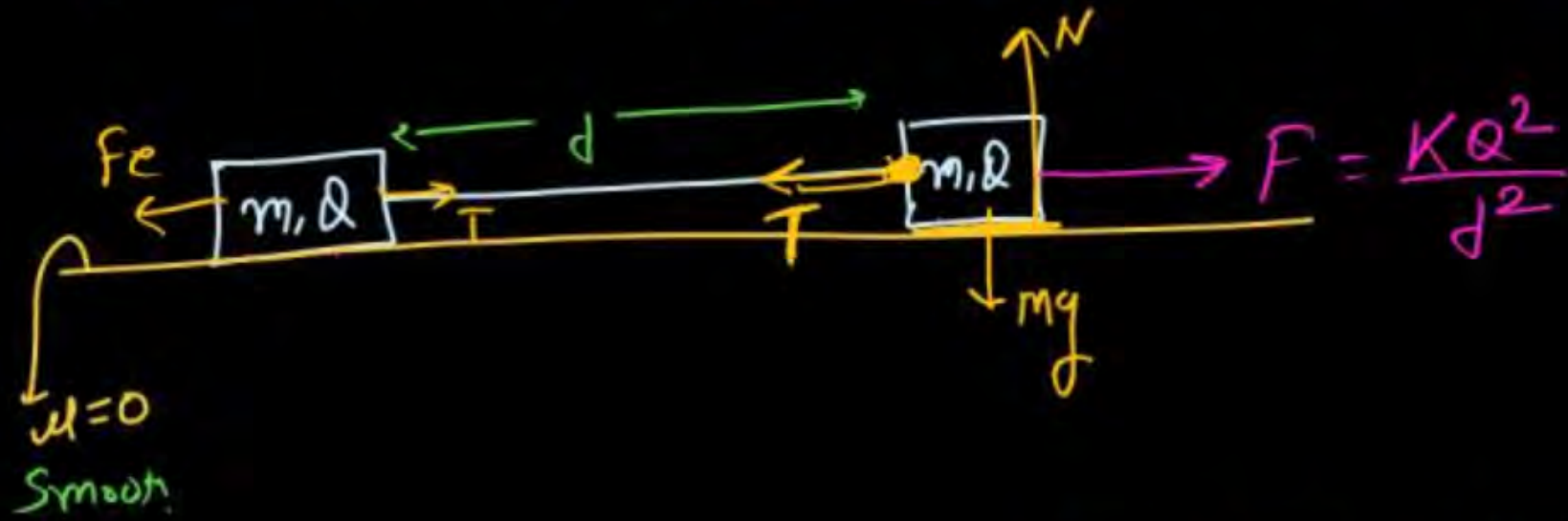
$$F = \frac{kq \times q}{x^2}$$



$$F' > F$$

Ans

(Q) if object at equilibrium then find tension in wire ??



(Copy)

net force in x-axis = 0

$$\frac{KQ^2}{d^2} = T$$

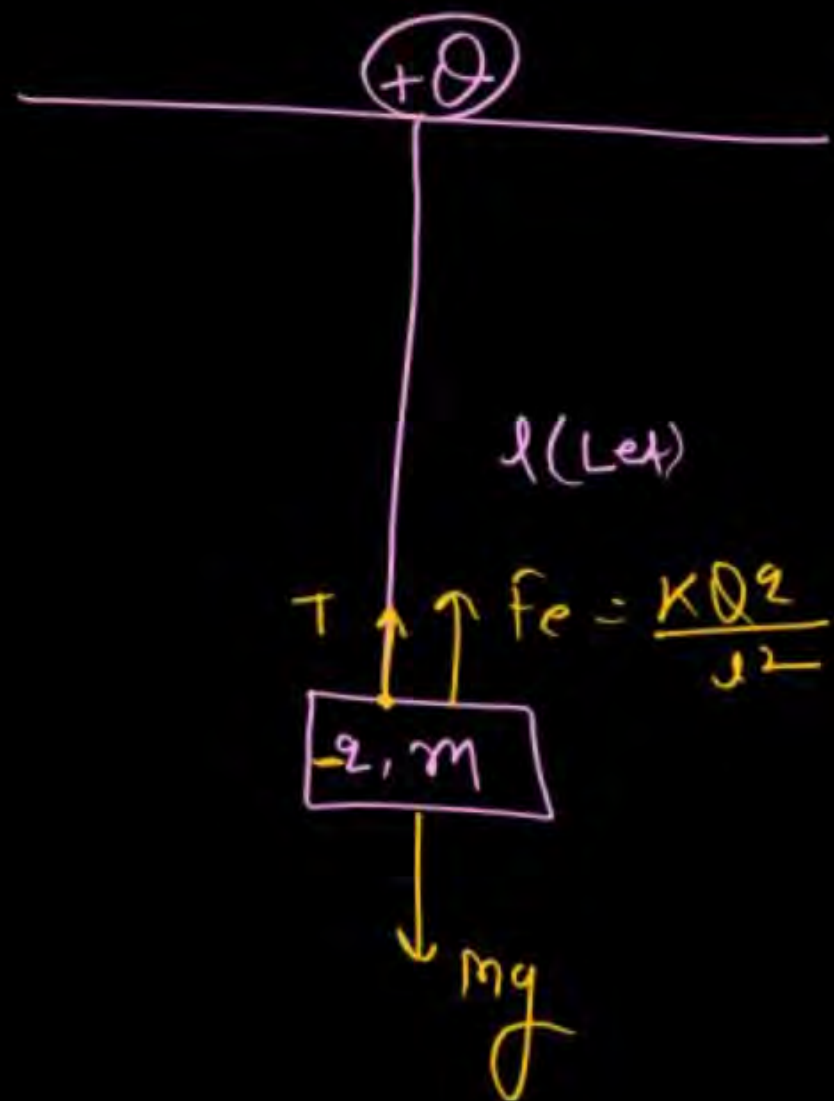
$$T = \frac{Q^2}{4\pi\epsilon_0 d^2}$$

⑧

find length of string, if
Tension in string is zero ??

Solⁿ

(or)



$$mg = \frac{kQq}{l^2}$$

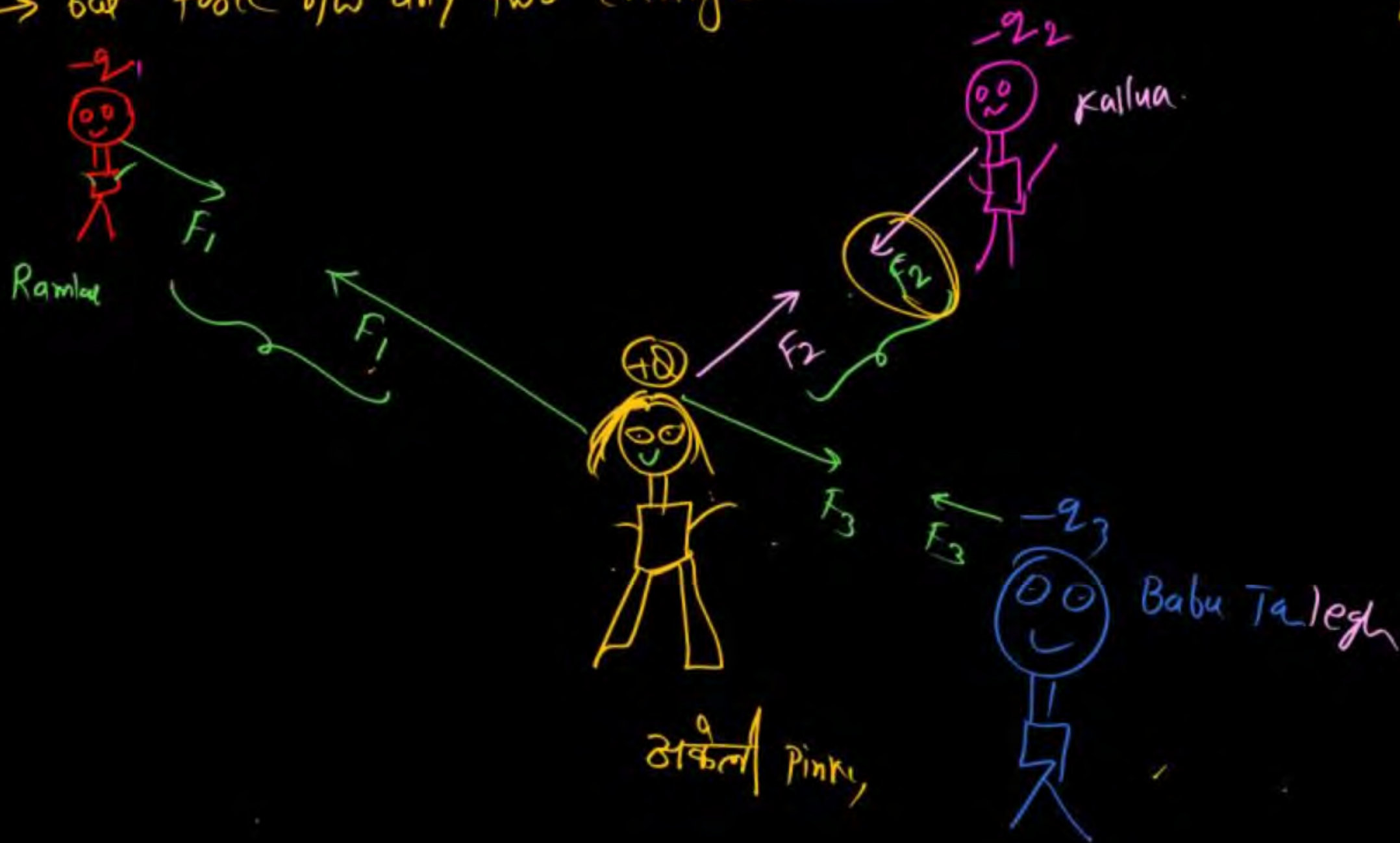
$$l = \sqrt{\frac{kQq}{mg}} \quad \underline{\underline{Ans}}$$

Superposition theorem $\rightarrow$ Net electrostatic force on any charge is vector sum of all the charge

$$F_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

$\rightarrow$ but force b/w any two charge does not depends on presence of other charge.

copy





@MRPHYSICSS

HOME WORK

- (1) Starting me 10 mint me Jo bhi MR sir ne bola hai
Sab ko rough page me likhna hai.
- (2) aat ke sare question again rough page pe karna hai
- (3) DO DPP
- (4) MR Physics Questions (46) to (67)



THANK
You